DETAIL CALCULATION BOOKLET

Vessel Name:	M. V. YOUNUS MOLLA-1	Operating Route:	Inland & bay crossing
Vessel Type:	Steel Body Cargo Vessel	Plying Time:	Day & Night
Hull Construction:	Single Hull	Statutory Factor kt / kz:	1.40 / 1.40
Length Over All (LOA):	76.21 m	Breadth Moulded (B):	15.24 m
Length BP (LBP):	69.85 m	Depth Moulded (D):	4.87 m
Design Draft (d):	4 m	Block Coeff (Cb):	0.721
Transverse Frame Spacing:	550 mm		

PREPARED & DESIGNED BY	CHECKED & VERIFIED BY
ROUNAK SAHA NILOY Naval Architect and Marine Engineer Designer ShipDyn Ltd.	Amitavo Chowdhury Wye Authorized Person ShipDyn Ltd.

EXECUTIVE STATUTORY COMPLIANCE DASHBOARD

Comprehensive Statutory Scantling Audit organized by 12 Statutory Structural Categories | Construction: Single Hull

GROUP 1: 1. HULL SHELL & DECK PLATING (15 items) — 15 / 15 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
1	Minimum Hull Plate Material	4.00	1.40	5.60	**6.00**	mm	+0.40	PASS
2	Keel Plating Thickness	7.99	1.40	11.19	**12.00**	mm	+0.81	PASS
3	Keel Plating Width	1099.20	1.00	1099.20	**1100.00**	mm	+0.80	PASS
4	Bottom Shell Plating	7.49	1.40	10.49	**12.00**	mm	+1.51	PASS
5	Bilge Plating Thickness	9.49	1.40	13.29	**14.00**	mm	+0.71	PASS
6	Side Shell Plating	7.49	1.40	10.49	**12.00**	mm	+1.51	PASS
7	Sheer Strake Plating	7.49	1.40	10.49	**12.00**	mm	+1.51	PASS
8	Main Strength Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
9	Poop Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
10	Forecastle Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
11	Second / Tween Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
12	Navigation Bridge Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
13	Strength Deck at Hatchways	8.99	1.40	12.59	**13.00**	mm	+0.41	PASS
14	Plate Stem Plating	10.98	1.40	15.38	**16.00**	mm	+0.62	PASS
15	Transom Stern Plating	7.49	1.40	10.49	**11.00**	mm	+0.51	PASS

GROUP 2: 2. STEMS, STERN FRAMES & RUDDER (16 items) — 16 / 16 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
16	Bar Stem Breadth & Thickness	164.80	1.00	164.80	**164.80**	mm	+0.00	PASS
17	Stern Frame Bar Breadth	164.80	1.00	164.80	**164.80**	mm	+0.00	PASS
18	Propeller Post Dimensions	274.60	1.00	274.60	**274.60**	mm	+0.00	PASS
19	Sole Piece Breadth	209.70	1.00	209.70	**209.70**	mm	+0.00	PASS
20	Sole Piece Thickness	74.90	1.00	74.90	**74.90**	mm	+0.00	PASS
21	Propeller Boss Wall Thickness	38.00	1.00	38.00	**38.00**	mm	+0.00	PASS
22	V-Strut Propeller Shaft Bracket	38.00	1.00	38.00	**38.00**	mm	+0.00	PASS
23	I-Strut Propeller Shaft Bracket	54.00	1.00	54.00	**54.00**	mm	+0.00	PASS
24	Rudder Stock Diameter - Supported	193.80	1.00	193.80	**193.80**	mm	+0.00	PASS
25	Rudder Stock Diameter - Spade	193.80	1.00	193.80	**193.80**	mm	+0.00	PASS
26	Rudder Pintle Diameter	193.80	1.00	193.80	**193.80**	mm	+0.00	PASS
27	Rudder Coupling Flange Thickness	30.20	1.00	30.20	**30.20**	mm	+0.00	PASS
28	Rudder Coupling Bolts Diameter	40.40	1.00	40.40	**40.40**	mm	+0.00	PASS
29	Rudder Blade Plating Thickness	7.49	1.40	10.49	**12.00**	mm	+1.51	PASS
30	Steering Tiller / Quadrant Hub	1304.00	1.00	1304.00	**1304.00**	mm	+0.00	PASS
31	Steering Chains & Shackles	16.00	1.00	16.00	**16.00**	mm	+0.00	PASS

GROUP 3: 3. BOTTOM CONSTRUCTION, FLOORS & KEELSONS (3 items) — 0 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
32	Centre Keelson	1057.77	1.40	1480.88	**1644.43**	cm3	+163.55	PASS
33	Side Keelson	400.00	1.40	560.00	**916.02**	cm3	+356.02	PASS
34	Bottom Transverse Floor Plate	2027.54	1.40	2838.56	**1239.57**	cm3	-1598.99	FAIL

GROUP 4: 4. FRAMING SYSTEMS & STRINGERS (5 items) — 5 / 5 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
35	Main Side Transverse Frame	199.40	1.40	279.16	**349.69**	cm3	+70.53	PASS
36	Side Web Frame	212.77	1.40	297.88	**471.50**	cm3	+173.62	PASS
37	Side Stringer	212.77	1.40	297.88	**432.49**	cm3	+134.61	PASS
38	Bilge Bracket	450.00	1.00	450.00	**600.00**	mm	+150.00	PASS
39	Deck Beam End Bracket to Frame	350.00	1.00	350.00	**450.00**	mm	+100.00	PASS

GROUP 5: 5. DECKS, GIRDERS & PILLARS (11 items) — 10 / 11 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
40	Solid Round Deck Pillar	57.00	1.00	57.00	**57.00**	mm	+0.00	PASS
41	Tubular Steel Deck Pillar	101.60	1.00	101.60	**101.60**	mm	+0.00	PASS
42	Transverse Strength Deck Beam	13.34	1.40	18.67	**128.14**	cm3	+109.47	PASS
43	Strength Deck Longitudinal Beams	13.34	1.40	18.67	**128.14**	cm3	+109.47	PASS
44	Strength Deck Transverse Web Beams	26.68	1.40	37.34	**813.65**	cm3	+776.31	PASS
45	Main Deck Center Longitudinal Girder	626.86	1.40	877.60	**813.65**	cm3	-63.95	FAIL
46	Superstructure Roof Beam	16.50	1.40	23.10	**86.83**	cm3	+63.73	PASS
47	Side Roof Girder	63.00	1.40	88.20	**182.20**	cm3	+94.00	PASS
48	Centre Roof Girder	80.00	1.40	112.00	**233.43**	cm3	+121.43	PASS
49	Superstructure Wall Stiffener	16.00	1.40	22.40	**71.84**	cm3	+49.44	PASS
50	Superstructure Main Frame	16.00	1.40	22.40	**118.29**	cm3	+95.89	PASS



GROUP 6: 6. WATERTIGHT BULKHEADS & TANKS (5 items) — 5 / 5 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
51	Collision Bulkhead Position	3.49	1.00	3.49	**4.80**	m	+1.31	PASS
52	Transverse Watertight Bulkhead Plating	5.50	1.40	7.70	**10.00**	mm	+2.30	PASS
53	Collision & Aft Peak Bulkhead Plating	6.00	1.40	8.40	**11.00**	mm	+2.60	PASS
54	Bow Chain Locker Wall Plating	6.98	1.00	6.98	**7.00**	mm	+0.02	PASS
55	Watertight Bulkhead Vertical Stiffeners	255.50	1.40	357.70	**370.93**	cm3	+13.23	PASS

GROUP 7: 7. HATCHWAYS, COAMINGS & COVERS (7 items) — 7 / 7 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
56	Longitudinal Hatch Coaming Plating	8.49	1.40	11.89	**13.00**	mm	+1.11	PASS
57	Transverse Hatch Coaming Plating	8.49	1.40	11.89	**13.00**	mm	+1.11	PASS
58	Hatch Coaming Stay Bracket	300.00	1.00	300.00	**300.00**	mm	+0.00	PASS
59	Cargo Hatchway Transverse Beams	13.00	1.40	18.20	**72.53**	cm3	+54.33	PASS
60	Cargo Hatchway Side Longitudinal Beams	13.00	1.40	18.20	**72.53**	cm3	+54.33	PASS
61	Portable Removable Hatch Beam	13.00	1.40	18.20	**68.80**	cm3	+50.60	PASS
62	Steel Hatch Cover Stiffeners	5.00	1.40	7.00	**55.46**	cm3	+48.46	PASS

GROUP 8: 8. MACHINERY CASINGS & SEATINGS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
63	Machinery Casing Plating	5.00	1.40	7.00	**7.00**	mm	+0.00	PASS
64	Engine Foundation Top Plate	19.00	1.40	26.60	**27.00**	mm	+0.40	PASS
65	Engine Foundation Longitudinal Girders	12.00	1.40	16.80	**18.00**	mm	+1.20	PASS

GROUP 9: 9. BULWARKS & FREEING PORTS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
66	Bulwark Minimum Height	1000.00	1.00	1000.00	**1000.00**	mm	+0.00	PASS
67	Bulwark Plating Thickness	5.99	1.00	5.99	**6.00**	mm	+0.01	PASS
68	Freeing Port Total Area per Side	4.54	1.00	4.54	**4.54**	m2	+0.00	PASS

GROUP 10: 10. EQUIPMENT NUMBER, ANCHORS & CABLES (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
69	Equipment Number Z Calculation	343.90	1.00	343.90	**343.90**	Z	+0.00	PASS
70	Bower Anchor Mass	597.00	1.00	597.00	**597.00**	kg	+0.00	PASS
71	Stud Link Chain Cable Total Length	275.00	1.00	275.00	**275.00**	m	+0.00	PASS

GROUP 11: 11. MACHINERY SHAFTING & BEARINGS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
72	Intermediate Shaft Diameter	85.00	1.00	85.00	**85.00**	mm	+0.00	PASS
73	Thrust Shaft Diameter	95.00	1.00	95.00	**95.00**	mm	+0.00	PASS
74	Propeller Shaft Diameter	105.00	1.00	105.00	**105.00**	mm	+0.00	PASS

GROUP 12: 12. PIPING & PUMPING SYSTEMS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
75	Main Bilge Suction Pipe Diameter	74.90	1.00	74.90	**74.90**	mm	+0.00	PASS
76	Branch Bilge Suction Pipe Diameter	50.00	1.00	50.00	**50.00**	mm	+0.00	PASS
77	Bilge & Ballast Pipe Wall Thickness	3.05	1.00	3.05	**3.05**	mm	+0.00	PASS



SHEET 01 : CENTRE KEELSON | T 400x250x12 mm

APPLICATION FORM: ITEM 31 (Rule 23)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	400.0	12.0	4800.0
Stiffener Flange	250.0	12.0	3000.0
Total Section Area			**14400.0**

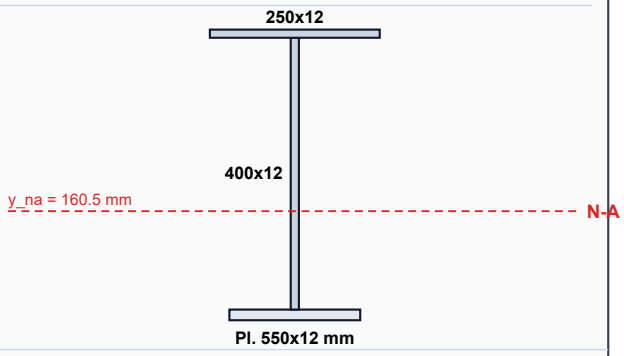
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	160.500 mm
Plate Parallel-Axis Inertia (Ip):	157622850.00 mm⁴
Web Parallel-Axis Inertia (Iw):	76730800.00 mm⁴
Flange Parallel-Axis Inertia (If):	198954750.00 mm⁴
Combined Section Inertia (I_NA):	**433308400.00 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**1644.43 cm³ (1644434 mm³)**
Application Form Reference:	Item 31: Kilsons & Webs (Centre girder)
Statutory Rule & Form Page:	Rule 23 (Table 16) | Page 11 of 24

CENTRE KEELSON

Profile: T 400x250x12 mm

Section Modulus: Z = 1644.43 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 02 : SIDE KEELSON | T 400x100x11 mm

APPLICATION FORM: ITEM 31 (Rule 23)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	400.0	11.0	4400.0
Stiffener Flange	100.0	11.0	1100.0
Total Section Area			**12100.0**

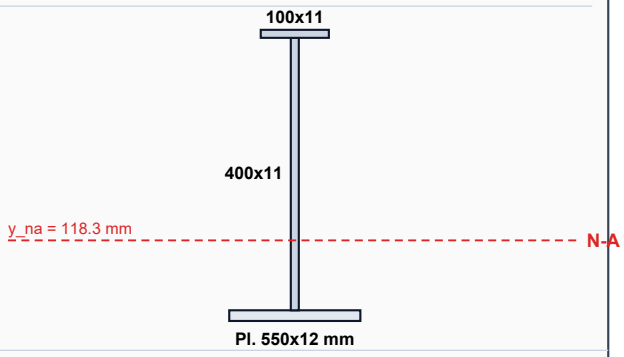
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	118.318 mm
Plate Parallel-Axis Inertia (Ip):	83340668.18 mm⁴
Web Parallel-Axis Inertia (Iw):	97282312.12 mm⁴
Flange Parallel-Axis Inertia (If):	98471828.03 mm⁴
Combined Section Inertia (I_NA):	**279094808.33 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**916.02 cm³ (916021 mm³)**
Application Form Reference:	Item 31: Kilsons & Webs (Side girder)
Statutory Rule & Form Page:	Rule 23 (Table 15) | Page 11 of 24

SIDE KEELSON

Profile: T 400x100x11 mm

Section Modulus: Z = 916.02 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 03 : BOTTOM TRANSVERSE FLOOR PLATE | T 400x180x11 mm

APPLICATION FORM: ITEM 28 (Rule 22)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	400.0	11.0	4400.0
Stiffener Flange	180.0	11.0	1980.0
Total Section Area			**12980.0**

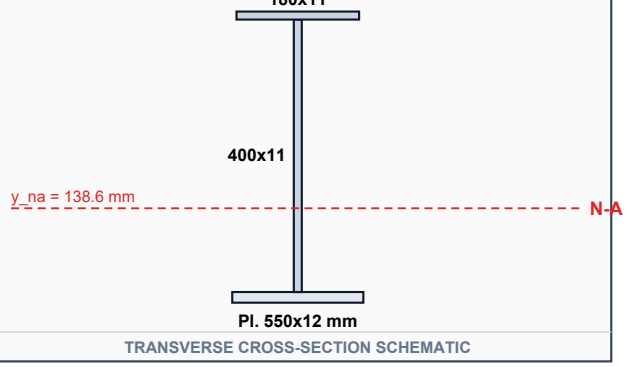
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	138.602 mm
Plate Parallel-Axis Inertia (Ip):	116128382.66 mm⁴
Web Parallel-Axis Inertia (Iw):	82370835.90 mm⁴
Flange Parallel-Axis Inertia (If):	154032808.87 mm⁴
Combined Section Inertia (I_NA):	**352532027.43 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**1239.57 cm³ (1239571 mm³)**
Application Form Reference:	Item 28: Bottom construction (Floor plate)
Statutory Rule & Form Page:	Rule 22 (Table 13) | Page 11 of 24

BOTTOM TRANSVERSE FLOOR PLATE

Profile: T 400x180x11 mm

Section Modulus: Z = 1239.57 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 04 : MAIN SIDE TRANSVERSE FRAME | L 200x100x11 mm

APPLICATION FORM: ITEM 33 (Rule 25)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	200.0	11.0	2200.0
Stiffener Flange	100.0	11.0	1100.0
Total Section Area			**9900.0**

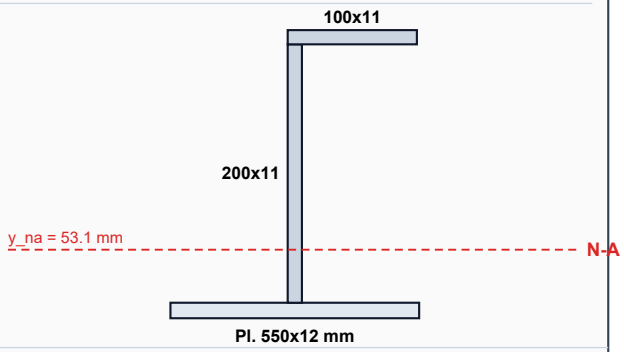
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	53.056 mm
Plate Parallel-Axis Inertia (Ip):	14693087.04 mm⁴
Web Parallel-Axis Inertia (Iw):	14977117.90 mm⁴
Flange Parallel-Axis Inertia (If):	29757264.51 mm⁴
Combined Section Inertia (I_NA):	**59427469.44 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**349.69 cm³ (349688 mm³)**
Application Form Reference:	Item 33: Side frame (Section modulus)
Statutory Rule & Form Page:	Rule 25 (Table 17) | Page 11 of 24

MAIN SIDE TRANSVERSE FRAME

Profile: L 200x100x11 mm

Section Modulus: Z = 349.69 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 05 : SIDE WEB FRAME | T 250x100x11 mm

APPLICATION FORM: ITEM 35 (Rule 26)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	250.0	11.0	2750.0
Stiffener Flange	100.0	11.0	1100.0
Total Section Area			**10450.0**

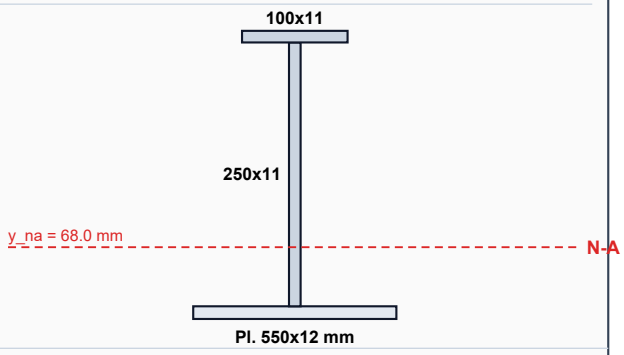
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	68.000 mm
Plate Parallel-Axis Inertia (Ip):	25449600.00 mm⁴
Web Parallel-Axis Inertia (Iw):	27415666.67 mm⁴
Flange Parallel-Axis Inertia (If):	43791366.67 mm⁴
Combined Section Inertia (I_NA):	**96656633.33 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**471.50 cm³ (471496 mm³)**
Application Form Reference:	Item 35: Web frame (Transverse framing)
Statutory Rule & Form Page:	Rule 26 (Table 19) | Page 12 of 24

SIDE WEB FRAME

Profile: T 250x100x11 mm

Section Modulus: Z = 471.50 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 06 : SIDE STRINGER | T 250x100x10 mm

APPLICATION FORM: ITEM 36 (Rule 27)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	250.0	10.0	2500.0
Stiffener Flange	100.0	10.0	1000.0
Total Section Area			**10100.0**

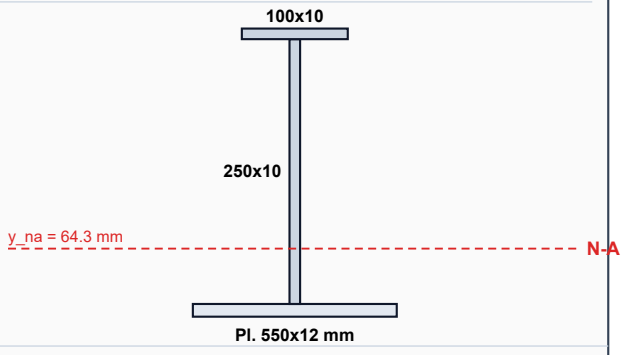
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	64.267 mm
Plate Parallel-Axis Inertia (Ip):	22486737.01 mm⁴
Web Parallel-Axis Inertia (Iw):	26245937.73 mm⁴
Flange Parallel-Axis Inertia (If):	41108870.14 mm⁴
Combined Section Inertia (I_NA):	**89841544.88 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**432.49 cm³ (432486 mm³)**
Application Form Reference:	Item 36: Side stringer (between web framing)
Statutory Rule & Form Page:	Rule 27 (Table 19) | Page 12 of 24

SIDE STRINGER

Profile: T 250x100x10 mm

Section Modulus: Z = 432.49 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 07 : TRANSVERSE STRENGTH DECK BEAM | L 125x75x9 mm

APPLICATION FORM: ITEM 37 (Rule 28)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	125.0	9.0	1125.0
Stiffener Flange	75.0	9.0	675.0
Total Section Area			**6750.0**

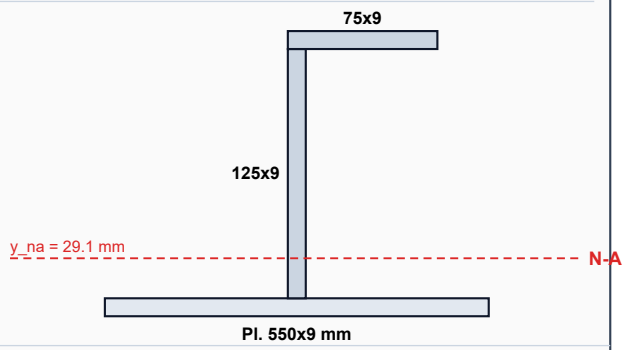
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	29.067 mm
Plate Parallel-Axis Inertia (Ip):	3020842.00 mm⁴
Web Parallel-Axis Inertia (Iw):	3490505.00 mm⁴
Flange Parallel-Axis Inertia (If):	8088123.00 mm⁴
Combined Section Inertia (I_NA):	**14599470.00 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**128.14 cm³ (128140 mm³)**
Application Form Reference:	Item 37: Deck beam (Section modulus)
Statutory Rule & Form Page:	Rule 28 (Table 20) | Page 12 of 24

TRANSVERSE STRENGTH DECK BEAM

Profile: L 125x75x9 mm

Section Modulus: Z = 128.14 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 08 : STRENGTH DECK LONGITUDINAL BEAMS | L 125x75x9 mm

APPLICATION FORM: ITEM 37 (Rule 28)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	125.0	9.0	1125.0
Stiffener Flange	75.0	9.0	675.0
Total Section Area			**6750.0**

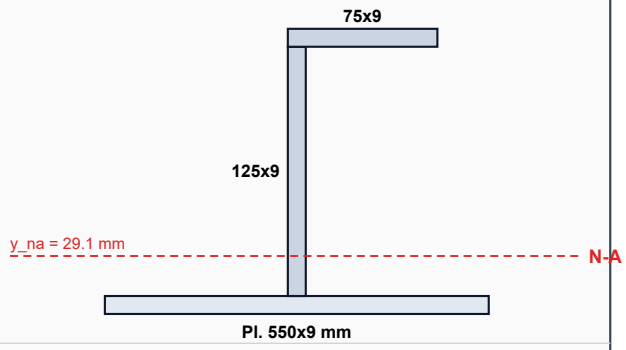
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	29.067 mm
Plate Parallel-Axis Inertia (Ip):	3020842.00 mm⁴
Web Parallel-Axis Inertia (Iw):	3490505.00 mm⁴
Flange Parallel-Axis Inertia (If):	8088123.00 mm⁴
Combined Section Inertia (I_NA):	**14599470.00 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**128.14 cm³ (128140 mm³)**
Application Form Reference:	Item 37: Strength deck longitudinal beam
Statutory Rule & Form Page:	Rule 28 (Table 20) | Page 12 of 24

STRENGTH DECK LONGITUDINAL BEAMS

Profile: L 125x75x9 mm

Section Modulus: Z = 128.14 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 09 : STRENGTH DECK TRANSVERSE WEB BEAMS | T 300x175x11 mm

APPLICATION FORM: ITEM 38 (Rule 29)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	300.0	11.0	3300.0
Stiffener Flange	175.0	11.0	1925.0
Total Section Area			**10175.0**

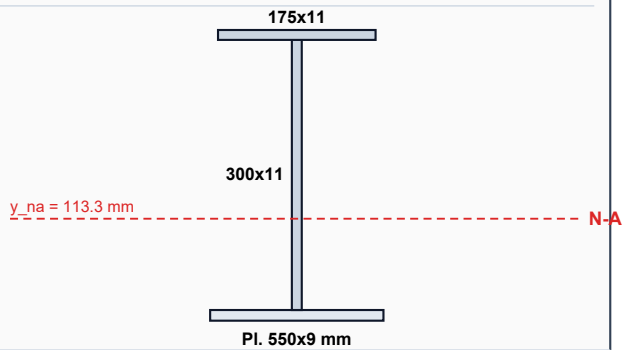
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	113.257 mm
Plate Parallel-Axis Inertia (Ip):	58582171.59 mm⁴
Web Parallel-Axis Inertia (Iw):	31655066.20 mm⁴
Flange Parallel-Axis Inertia (If):	77979683.10 mm⁴
Combined Section Inertia (I_NA):	**168216920.89 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**813.65 cm³ (813651 mm³)**
Application Form Reference:	Item 38: Deck web beam (Section modulus)
Statutory Rule & Form Page:	Rule 29 (Rule 29, Table 20) | Page 12 of 24

STRENGTH DECK TRANSVERSE WEB B..

Profile: T 300x175x11 mm

Section Modulus: Z = 813.65 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
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SHEET 10 : MAIN DECK CENTER LONGITUDINAL GIRDER | T 300x175x11 mm

APPLICATION FORM: ITEM 39 (Rule 30)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	300.0	11.0	3300.0
Stiffener Flange	175.0	11.0	1925.0
Total Section Area			**10175.0**

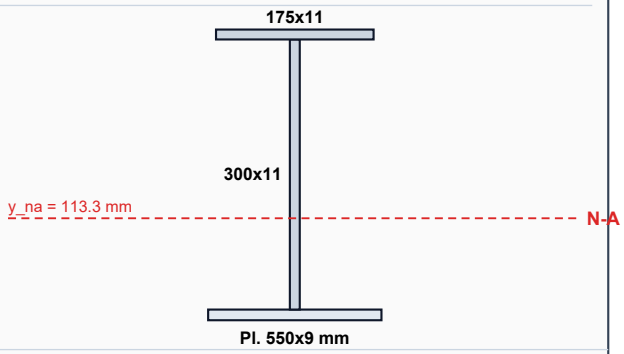
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	113.257 mm
Plate Parallel-Axis Inertia (Ip):	58582171.59 mm⁴
Web Parallel-Axis Inertia (Iw):	31655066.20 mm⁴
Flange Parallel-Axis Inertia (If):	77979683.10 mm⁴
Combined Section Inertia (I_NA):	**168216920.89 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**813.65 cm³ (813651 mm³)**
Application Form Reference:	Item 39: Deck girder (Section modulus)
Statutory Rule & Form Page:	Rule 30 (Table 21) | Page 12 of 24

MAIN DECK CENTER LONGITUDINAL ..

Profile: T 300x175x11 mm

Section Modulus: Z = 813.65 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 11 : SUPERSTRUCTURE ROOF BEAM | L 100x75x8 mm

APPLICATION FORM: ITEM 74 (Rule 63)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	100.0	8.0	800.0
Stiffener Flange	75.0	8.0	600.0
Total Section Area			**6350.0**

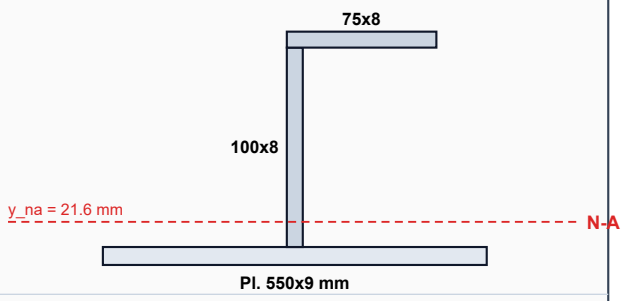
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	21.618 mm
Plate Parallel-Axis Inertia (Ip):	1483909.51 mm⁴
Web Parallel-Axis Inertia (Iw):	1784591.21 mm⁴
Flange Parallel-Axis Inertia (If):	5013589.87 mm⁴
Combined Section Inertia (I_NA):	**8282090.58 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**86.83 cm³ (86831 mm³)**
Application Form Reference:	Item 74: Superstructure roof beam
Statutory Rule & Form Page:	Rule 63 (Table 36) | Page 17 of 24

SUPERSTRUCTURE ROOF BEAM

Profile: L 100x75x8 mm

Section Modulus: Z = 86.83 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 12 : SIDE ROOF GIRDER | L 150x90x9 mm

APPLICATION FORM: ITEM 74 (Rule 63)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	150.0	9.0	1350.0
Stiffener Flange	90.0	9.0	810.0
Total Section Area			**7110.0**

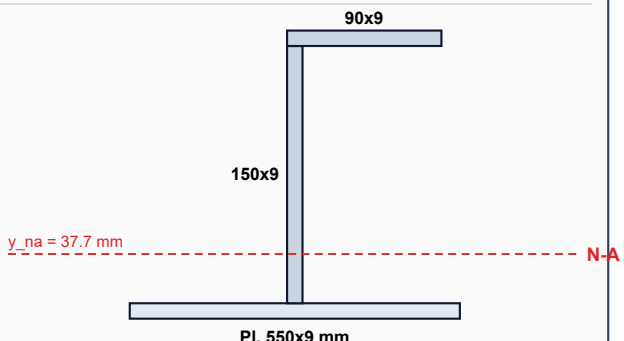
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	37.709 mm
Plate Parallel-Axis Inertia (Ip):	5492413.24 mm⁴
Web Parallel-Axis Inertia (Iw):	5424123.92 mm⁴
Flange Parallel-Axis Inertia (If):	12822430.18 mm⁴
Combined Section Inertia (I_NA):	**23738967.34 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**182.20 cm³ (182199 mm³)**
Application Form Reference:	Item 74: Superstructure side roof girder
Statutory Rule & Form Page:	Rule 63 (Table 37) | Page 17 of 24

SIDE ROOF GIRDER

Profile: L 150x90x9 mm

Section Modulus: Z = 182.20 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

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SHEET 13 : CENTRE ROOF GIRDER | L 160x100x10 mm

APPLICATION FORM: ITEM 74 (Rule 63)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	160.0	10.0	1600.0
Stiffener Flange	100.0	10.0	1000.0
Total Section Area			**7550.0**

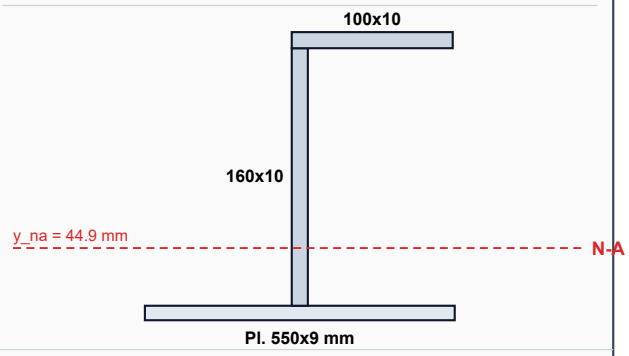
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	44.858 mm
Plate Parallel-Axis Inertia (Ip):	8095661.45 mm⁴
Web Parallel-Axis Inertia (Iw):	6531013.45 mm⁴
Flange Parallel-Axis Inertia (If):	16686088.71 mm⁴
Combined Section Inertia (I_NA):	**31312763.60 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**233.43 cm³ (233429 mm³)**
Application Form Reference:	Item 74: Superstructure centre roof girder
Statutory Rule & Form Page:	Rule 63 (Table 37) | Page 17 of 24

CENTRE ROOF GIRDER

Profile: L 160x100x10 mm

Section Modulus: Z = 233.43 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 14 : SUPERSTRUCTURE WALL STIFFENER | L 100x65x7 mm

APPLICATION FORM: ITEM 74 (Rule 63)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	100.0	7.0	700.0
Stiffener Flange	65.0	7.0	455.0
Total Section Area			**7755.0**

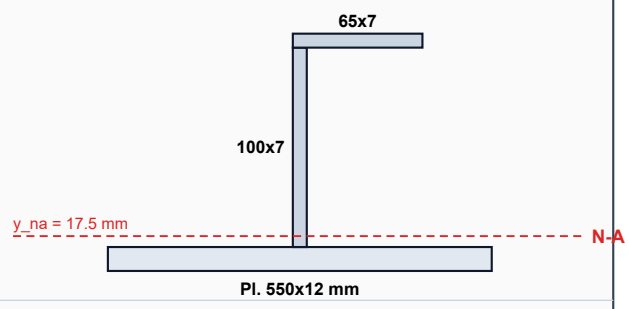
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	17.479 mm
Plate Parallel-Axis Inertia (Ip):	948920.89 mm⁴
Web Parallel-Axis Inertia (Iw):	1970794.00 mm⁴
Flange Parallel-Axis Inertia (If):	4373518.06 mm⁴
Combined Section Inertia (I_NA):	**7293232.95 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**71.84 cm³ (71840 mm³)**
Application Form Reference:	Item 74: Superstructure wall stiffener
Statutory Rule & Form Page:	Rule 63 (Table 35) | Page 17 of 24

SUPERSTRUCTURE WALL STIFFENER

Profile: L 100x65x7 mm

Section Modulus: Z = 71.84 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 15 : SUPERSTRUCTURE MAIN FRAME | L 125x75x8 mm

APPLICATION FORM: ITEM 74 (Rule 63)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	125.0	8.0	1000.0
Stiffener Flange	75.0	8.0	600.0
Total Section Area			**8200.0**

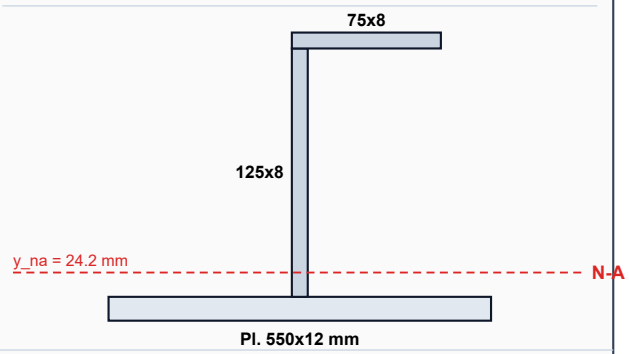
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	24.232 mm
Plate Parallel-Axis Inertia (Ip):	2273008.00 mm⁴
Web Parallel-Axis Inertia (Iw):	3828984.58 mm⁴
Flange Parallel-Axis Inertia (If):	8184100.51 mm⁴
Combined Section Inertia (I_NA):	**14286093.09 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**118.29 cm³ (118293 mm³)**
Application Form Reference:	Item 74: Superstructure main frame
Statutory Rule & Form Page:	Rule 63 (Table 35) | Page 17 of 24

SUPERSTRUCTURE MAIN FRAME

Profile: L 125x75x8 mm

Section Modulus: Z = 118.29 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

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SHEET 16 : WATERTIGHT BULKHEAD VERTICAL STIFFENERS | L 200x100x12 mm

APPLICATION FORM: ITEM 44 (Rule 34)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	10.0	5500.0
Stiffener Web	200.0	12.0	2400.0
Stiffener Flange	100.0	12.0	1200.0
Total Section Area			**9100.0**

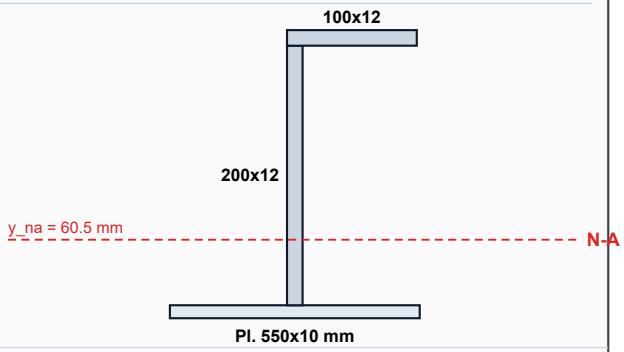
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	60.516 mm
Plate Parallel-Axis Inertia (Ip):	16997273.01 mm⁴
Web Parallel-Axis Inertia (Iw):	13876684.17 mm⁴
Flange Parallel-Axis Inertia (If):	29024548.68 mm⁴
Combined Section Inertia (I_NA):	**59898505.86 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**370.93 cm³ (370926 mm³)**
Application Form Reference:	Item 44: Watertight bulkhead stiffeners
Statutory Rule & Form Page:	Rule 34 (Table 26) | Page 13 of 24

WATERTIGHT BULKHEAD VERTICAL S..

Profile: L 200x100x12 mm

Section Modulus: Z = 370.93 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 17 : CARGO HATCHWAY TRANSVERSE BEAMS | L 75x75x9 mm

APPLICATION FORM: ITEM 60 (Rule 50)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	13.0	7150.0
Stiffener Web	75.0	9.0	675.0
Stiffener Flange	75.0	9.0	675.0
Total Section Area			**8500.0**

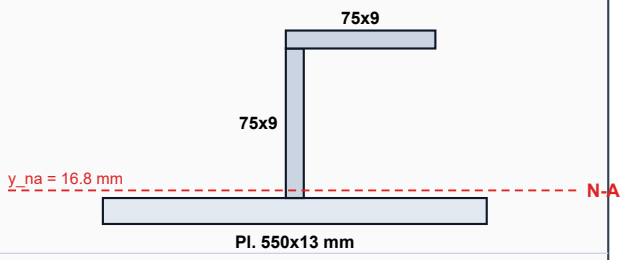
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	16.824 mm
Plate Parallel-Axis Inertia (Ip):	862708.94 mm⁴
Web Parallel-Axis Inertia (Iw):	1081926.90 mm⁴
Flange Parallel-Axis Inertia (If):	3870232.79 mm⁴
Combined Section Inertia (I_NA):	**5814868.63 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**72.53 cm³ (72526 mm³)**
Application Form Reference:	Item 60: Hatchway transverse beam
Statutory Rule & Form Page:	Rule 50 (Table 32) | Page 15 of 24

CARGO HATCHWAY TRANSVERSE BEAMS

Profile: L 75x75x9 mm

Section Modulus: Z = 72.53 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 18 : CARGO HATCHWAY SIDE LONGITUDINAL BEAMS | T 75x75x9 mm

APPLICATION FORM: ITEM 62 (Rule 51)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	13.0	7150.0
Stiffener Web	75.0	9.0	675.0
Stiffener Flange	75.0	9.0	675.0
Total Section Area			**8500.0**

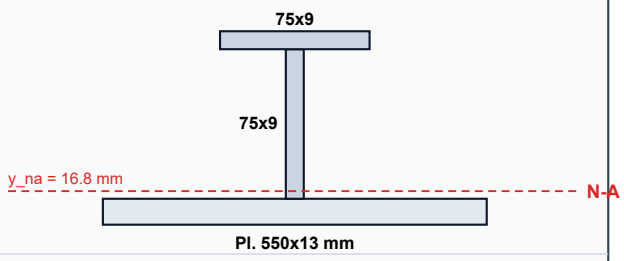
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	16.824 mm
Plate Parallel-Axis Inertia (Ip):	862708.94 mm⁴
Web Parallel-Axis Inertia (Iw):	1081926.90 mm⁴
Flange Parallel-Axis Inertia (If):	3870232.79 mm⁴
Combined Section Inertia (I_NA):	**5814868.63 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**72.53 cm³ (72526 mm³)**
Application Form Reference:	Item 62: Hatchway side girder beam
Statutory Rule & Form Page:	Rule 51 (Table 32) | Page 15 of 24

CARGO HATCHWAY SIDE LONGITUDIN..

Profile: T 75x75x9 mm

Section Modulus: Z = 72.53 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 19 : PORTABLE REMOVABLE HATCH BEAM | T 75x75x9 mm

APPLICATION FORM: ITEM 60 (Rule 50)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	75.0	9.0	675.0
Stiffener Flange	75.0	9.0	675.0
Total Section Area			**6300.0**

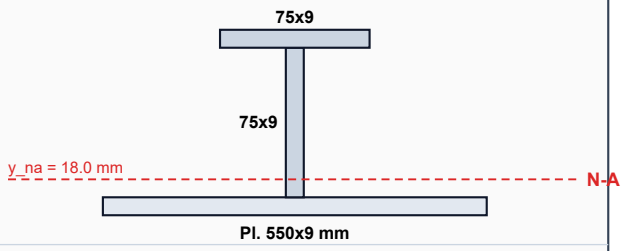
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	18.000 mm
Plate Parallel-Axis Inertia (Ip):	935550.00 mm⁴
Web Parallel-Axis Inertia (Iw):	864675.00 mm⁴
Flange Parallel-Axis Inertia (If):	3359475.00 mm⁴
Combined Section Inertia (I_NA):	**5159700.00 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**68.80 cm³ (68796 mm³)**
Application Form Reference:	Item 60: Portable removable hatch beam
Statutory Rule & Form Page:	Rule 50 (Table 32) | Page 15 of 24

PORTABLE REMOVABLE HATCH BEAM

Profile: T 75x75x9 mm

Section Modulus: Z = 68.80 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 20 : STEEL HATCH COVER STIFFENERS | L 75x65x8 mm

APPLICATION FORM: ITEM 64 (Rule 52)

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	75.0	8.0	600.0
Stiffener Flange	65.0	8.0	520.0
Total Section Area			**6070.0**

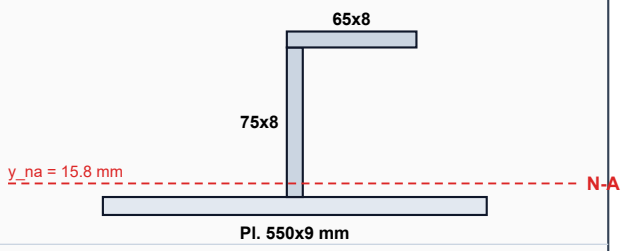
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	15.805 mm
Plate Parallel-Axis Inertia (Ip):	666012.58 mm⁴
Web Parallel-Axis Inertia (Iw):	846568.01 mm⁴
Flange Parallel-Axis Inertia (If):	2713091.41 mm⁴
Combined Section Inertia (I_NA):	**4225671.99 mm⁴**

CALCULATED SECTION MODULUS	
Section Modulus, Z:	**55.46 cm³ (55458 mm³)**
Application Form Reference:	Item 64: Steel hatch cover stiffeners
Statutory Rule & Form Page:	Rule 52 (Table 34) | Page 16 of 24

STEEL HATCH COVER STIFFENERS

Profile: L 75x65x8 mm

Section Modulus: Z = 55.46 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

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